

Session-03: Continuous random variables

1. The proportion of people who respond to a mail order solicitation is a continuous random variable X that has a density function given by:

$$f(x) = \begin{cases} \frac{2(x+2)}{5} & \text{if } 0 < X < 1 \\ 0 & \text{otherwise} \end{cases}$$

- a) Show that $P(0 < X < 1) = 1$.

(Hint: Show that the given function integrates to 1 between the limits 0 and 1.)

- b) Find the probability that more than $\frac{1}{4}$ but fewer than $\frac{1}{2}$ of the people contacted will respond to this type of solicitation.

(Hint: Integrate the given function between the limits $1/4$ and $1/2$.)

2. The pdf of the samples of speech waveforms is found to decay exponentially at a rate α , so that the following pdf is proposed:

$$f(x) = c e^{\alpha|x|} \quad -\infty < X < \infty$$

Find the constant C and then find the probability $P(|X| < v)$.

3. The CDF of checkout time duration X is $F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^2}{4} & \text{if } 0 \leq x < 2, \\ 1 & \text{if } x \geq 2 \end{cases}$

Use this information to compute a) $P(X \leq 1)$ b) $P(0.5 \leq X \leq 1)$ c) Find the density function of X .

(Hint: Derivative of the CDF is the density function. The area under the density curve in an interval gives the probabilities.)